

GROUP 3 Members:

- 1)Aryan Rana
- 2)Omkar Bodake
- 3) A. Pavan Sri Sai
- 4) Sai Desale
- 5)Aman Chourasia

Week 3	Member 1 – Skid Buffer Design	Studied AXI VALID–READY handshake in detail and analyzed timing issues in combinational READY paths. Designed RTL architecture of the skid buffer and implemented temporary data holding logic to handle downstream stalls.
	Member 2 – Address Decoder	Understood AXI address mapping concepts and defined slave address ranges. Designed decoder logic using base address and mask comparison and generated one-hot slave select signals.
	Member 3 – Arbitration & Control Logic	Analyzed multi-master to single-slave access scenarios. Designed basic request–grant arbitration structure and studied priority handling mechanisms.
	Member 4 – Arbitration & Control Logic	Worked on control flow for transaction management. Studied coordination between write and read channels and helped define control signals required for grant holding and release.
	Member 5 – Request Tracking	Studied AXI-Lite transaction flow and response behavior. Analyzed how requests need to be tracked for correct response routing and began defining tracking signal requirements.

Week 4	Member 1 – Skid Buffer Design	Assisted in integration planning of skid buffer with arbitration block. Verified stall and resume behavior under different READY conditions.
	Member 2 – Address Decoder	Validated address decoding outputs with different test address ranges. Supported arbitration team by providing stable slave select signals.
	Member 3 – Arbitration & Control Logic	Implemented detailed request–grant logic and priority selection. Developed slave busy detection and grant generation mechanism.
	Member 4 – Arbitration & Control Logic	Designed control logic for grant hold until transaction completion. Worked on coordination between channel states and arbitration outputs.
	Member 5 – Request Tracking	Defined structure for tracking active transactions. Studied response channel timing and prepared logic plan for mapping responses back to requesting masters.
Basic Verification	Member 1 and 2	Continued simulation of input interface blocks with extended scenarios. Checked behavior under backpressure and invalid address conditions.